Goel *et al. Discover Applied Sciences* (2025) 7:1165
https://doi.org/10.1007/s42452-025-07728-1

Discover Applied Sciences

RESEARCH Open Access

Exploring automated machine learning to develop facial expression recognition systems

Atharv Goel[1], Praneet Saurabh[1*], Dhananjay Bisen[2], Rishav Dubey[1] and Prathap Somu[1]

*Correspondence:
Praneet Saurabh
praneet.saurabh@jaipur.manipal.edu
[1]Manipal University Jaipur, Jaipur, Rajasthan, India
[2]Madhav Institute of Technology and Science, Gwalior, M.P, India

Abstract

With human-computer interactions on the rise, in use for even the most mundane things like shopping, making travel arrangements and ordering food as well as matters of national importance such as healthcare and security, Facial Expression Recognition has become crucial to our lives. Currently used traditional facial expression recognition systems rely on manual tuning by experts to get optimised results, face challenges such as the demand of resources and expert-level knowledge of a plethora of neural network architectures which prevents the technology from being widely adopted. This paper demonstrates a practical application of Automated Machine Learning using a framework known as AutoGluon to automate model selection and hyperparameter tuning for Facial Expression Recognition, achieving competitive accuracy (76.4%) on the challenging FER2013 dataset. It also provides a comprehensive analysis of four model configurations (Default, Low Resource, Balanced, High Quality), enabling users to balance computational cost and performance. These models outperform several state-of-the-art manually tuned models (RMN, ViT, MLFCC etc.) and highlight AutoML's potential to democratize FER development for non-experts. This paper also includes the use of metrics such as precision, recall and f1-score along with accuracy to provide a more holistic understanding of the models' strengths and limitations, and includes ethical considerations and future directions, addressing critical issues like bias mitigation and real-world deployment challenges. This work confirms the potential of AutoML to revolutionize facial expression recognition, making advanced facial expression analysis more accessible and effective.

Article highlights

- A novel model with automated model selection, hyperparameter tuning and ensembling achieves state-of-the-art accuracy of 76.4% on the FER2013 dataset, surpassing several manually tuned models.
- Evaluates four model configurations, namely Default, Low Resource, Balanced, and High Quality, allowing users to balance computational cost and accuracy based on available resources.

© The Author(s) 2025. **Open Access** This article is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License, which permits any non-commercial use, sharing, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if you modified the licensed material. You do not have permission under this licence to share adapted material derived from this article or parts of it. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit http://creativecommons.org/licenses/by-nc-nd/4.0/.

- Discusses potential ethical challenges in FER applications and proposes future research directions, including cross-dataset generalization, model interpretability, and bias mitigation.

Keywords Facial expression recognition, Automated machine learning, AutoGluon, FER2013

1 Introduction

Facial expressions have always been an elemental aspect of communication, conveying a wealth of emotional and social information. From joy and sadness to anger and surprise, facial expressions are a major aspect of interpersonal interactions, affective computing, and human-computer interaction. Accurately interpreting facial expressions is crucial to various domains, including psychology, medicine, marketing, security, and entertainment. Recent advances in machine learning (ML) have transformed the field of facial expression recognition (FER), enabling effective real-world applications, particularly in the context of big data [1, 2]. Some of these applications are:

(i) Emotion analysis in large-scale market research.
(ii) Monitoring patients in healthcare department.
(iii) Real-time surveillance of public areas for suspicious behaviour.
(iv) Assessment of student engagement in online learning.
(v) Detecting driver drowsiness to improve road safety.

More specifically, FER in healthcare [3] is of utmost importance as demonstrated through the revolutionary company PainChek, which uses AI-driven facial analysis to assess pain levels in patients who cannot communicate (such as those with dementia), where the AI analyses micro-expressions associated with pain and generates an objective pain score. This allows for timely pain relief in residents who might otherwise suffer silently [4]. Another application of FER systems in healthcare was at the Yokohama City University Hospital, which developed an AI system to watch over sedated intensive care patients by looking for subtle signs of discomfort in the patients' faces and eyes [5].

FER, once solely a human domain, has been revolutionized by artificial intelligence (AI), enabling machines to process vast image and video data to detect and interpret expressions. As technology advances, FER plays a vital role in enhancing human-computer interaction by enabling systems to recognize emotions and intentions. Given that facial expressions convey 55% of the information being passed, with actual language constituting only 7% [6], FER offers powerful insights into a person's emotional and mental state. Its applications span virtual assistants, emotion aware gaming, personalized e-commerce experiences and more- making interfaces more intuitive, responsive and human centric.

With the rise of the internet, abundant image data is now available to train models for emotion recognition. Despite advances in FER technology [7], its real-world adoption remains limited due to persistent challenges. These include cultural differences in interpreting expressions [8], and inconsistencies in image quality, such as variations in lighting, occlusions, and resolution—common issues in Big Data. Overcoming these requires robust machine learning algorithms capable of generalizing across diverse conditions. Another key obstacle is the substantial resource demand—time, expertise, and computation—needed to design and optimize high-performing FER models. Even skilled data

scientists must invest significant effort in model selection, hyperparameter tuning, and training. This is where Automated Machine Learning (AutoML) becomes valuable [9] AutoML automates the end-to-end ML pipeline from raw data to model deployment using frameworks like AutoGluon, Google Cloud AutoML, and TPOT [10]. These tools simplify tasks such as model selection, hyperparameter optimization, feature engineering, and ensembling, making advanced model development accessible to non-experts. By automating complex and time-consuming processes, AutoML enables researchers to focus on higher-level insights, while efficiently identifying optimal model configurations through sophisticated search strategies.

In the context of FER, AutoML offers several advantages and opportunities for advancing the current state-of-the-art. Firstly, AutoML enables researchers to explore a plethora of ML models and architectures, including many foundational FER algorithms as well as convolutional neural networks (CNNs), recurrent neural networks (RNNs) and ensemble methods, without requiring specialized expertise in each algorithm. This flexibility allows for the identification of the most suitable model architecture for the given task and dataset, potentially leading to improved performance and generalization. Furthermore, AutoML frameworks facilitate efficient hyperparameter optimization, enabling the fine-tuning of model parameters to enhance performance and robustness. Hyperparameters play a critical role in determining a model's effectiveness, and AutoML techniques automate the process of searching for optimal hyperparameter configurations, significantly reducing the need for manual experimentation and tuning. This is done by employing optimization algorithms like Bayesian optimization and evolutionary algorithms.

In this paper, AutoGluon is employed to bridge the gap between experts and non-experts in the creation of models for FER. AutoGluon is used to automate the complex tasks of selecting models, adjusting the layers in the neural networks and optimizing hyperparameters and help the user create state-of-the-art models which are competitive to those created by experts. The proposed AutoGluon-driven method attempts to cover the underexplored area of AutoML in FER and succeeds in creating several models, which outperform the conventional state-of-the-art in accuracy on the FER2013 dataset [11]. The proposed AutoGluon model reported an accuracy of 76.4% on FER2013 dataset, higher than the traditional manually tuned models, which also shows its efficiency when processing Big Data. This work demonstrates that AutoML has the potential to improve FER systems, making advanced facial expression analysis ubiquitous and easy-to-use.

This paper is organised in the following way: Sect. 2 covers related work while Sect. 3 discusses the proposed work. Section 4 presents results, and its subsequent discussion and Sect. 5 concludes the paper.

2 Related work

Facial Expression Recognition (FER) is a computer vision function aimed at categorizing emotional expressions. The focus of this task is to automate the detection of facial expressions and thus the underlying emotional state by analysing various features and mapping them to emotions like anger, happiness, and fear.

Artificial Intelligence (AI) is a computer program which can compute or "think" in the way a human mind does. It is the ability of a program to transcend the limitations of

a regular program which is limited to computing exactly according to what the developer instructed it to, and instead creating its own mathematical structures and functions to provide an output based on prior information given to it [12, 13]. Machine learning (ML) [14] is a subset of AI, the ability given to a machine to learn and improve its performance without the need for a human to explain how to accomplish the task [15]. It does this by creating its own mathematical model based on patterns it identifies in data. In this field, there have been recent innovations to use ML for skin cancer lesion analysis, thus making ML an important part of the healthcare industry [16]. Deep learning (DL) [2] is a subset of ML where computer scientists create "neural networks," essentially layers of interconnected nodes or "neurons" to mimic the way the human mind processes information. A neural network consists of such a web of neurons, each of which has a mathematical algorithm (known as an activation function) to compute the information passed to it by the previous layer and its own weights and biases to provide an output. DL is thus the most efficient and intricate model, and it has the capability to work with tabular data, images and recreate human language in the form of text and even speech. Continuing the example, DL has also been used to further enhance the work of skin cancer diagnosis to improve the accuracy of the system [17].

FER systems cannot be handled by regular ML algorithms like Decision Trees and Regression because of the difficulties in identifying relevant features in images. These traditional ML algorithms are effective when the input features are explicitly defined and inherently meaningful, like in the case of numerical data, so for them to be of effective, the attributes of the image (edges and shapes) must be manually engineered and extracted into numerical representations before the algorithms can work with them. Raw image data alone do not carry such meaningful information. However, DL algorithms learn the hierarchical representations of images, with each layer extracting increasing abstract features from the input images. This allows the algorithm to capture complex patterns and relationships in the data, leading to superior classification performance. Therefore, DL is the best way to perform facial expression recognition.

2.1 Review of literature

This subsection discusses the various models designed by researchers to automate FER which build the expansive state-of-the-art which include CNNs, Attention-Based Models, and Vision transformers. Saeed et al. [6] developed an automated framework for facial detection, which they termed FD-CNN. The architecture consists of four convolutional layers and two hidden layers. The model is comprised of three processes; preprocessing, feature extraction, and classification. The CK + dataset is applied for both training and testing. Zhang et al. [18] introduced a novel approach dubbed Dual-Direction Attention Mixed Feature Network (DDAMFN). The model enhances accuracy with the use of dual-direction attention mechanisms (which enables attention to both fine details and broader context) alongside mixed feature fusion. The model maintained 67.03% accuracy with the AffectNet dataset, 90.74% with FER+, and 91.35% overall in RAF-DB. Later, Pham et al. [19] introduced a new approach for FER based on a Residual Masking Network (RMN) which enhances the performance of CNN. The architecture aims to achieve global and local feature capture. The architecture of the network comprises of residual blocks which capture image details on an intricate level. Moreover, it utilizes masking layers to both suppress irrelevant and enhance crucial discriminating

features. The RMN captures relevant information from the input image through the use of a segmentation network that partitions an image into several parts depending on the class to which they belong. On the FER2013 dataset, RMN was able to surpass the previous benchmark with a dipatched accuracy of 74.14% and VEMO's private dataset yielding a 65.94%. Later, Krizhevsky et al. [20] devised a Deep Convolutional Neural Network aimed for classifying the ImageNet database which harbours 1.2 million images partitioned into 1000 distinct categories. Their final model comprises 5 convolutional layers alongside 3 fully connected ones, employing a regularization technique known as 'dropout' to mitigate overfitting. This model was presented at the ImageNet ILSVRC-2012 contest, winning first place with 15.3% error rate against the runner up's 26.2%. Later, Jain et al. [21] introduced a model relying on a single Deep Convolutional Neural Network (DNN) featuring convolutional layers and deep residual blocks. The authors argue that incorporating residual blocks with CNN improves the results. The model was trained on the CK + and JAFFE datasets, achieving an accuracy of 93.24% and 95.23%, respectively.

On the other hand, Lopes et al. [22] 2022 developed a convolutional neural network (CNN)-based model for facial expression recognition (FER) with specific image preprocessing techniques. It is claimed that these methods are competitive against other approaches in the field of FER. These methods include the following: rotation correction, cropping, down-sampling, and intensity normalization. The model was evaluated using the CK + and JAFFE and BV-3DFE datasets and achieved the highest accuracy of 96.76% on CK+. Wang et al. [23] suggested a novel approach to FER based on the use of auxiliary models. The hypothesis is that using the entire face as input would result in a wealth of extraneous data which would lead to the loss of important features. This issue is mitigated through the use of multiple subregions of the face which are weighted relative to the rest of the image. The model was tested on four datasets: JAFFE, CK+, FER2013, and SFEW. Li et al. [24] proposed a new approach of FER aimed at enhancing human-robot interaction by using a CNN for information extraction, long short-term memory recurrent neural networks (LSTMs) for modelling the relation of facial expression changes over time with expressions in an image sequence and improved the model's performance with transfer learning. The model in question was evaluated using the JAFFE, CK+, FER2013, and SFEW datasets, obtaining accuracy values of 95.9%, 99.09%, 67.7%, and 59.97%, respectively. Later, Thacker [25] introduced three approaches to FER: Multi-Layer Feature Fusion based Classification (MLFFC), Multi-Model Feature Fusion based Classification (MMFFC), and a new model, EfficientNetB7, based on Normalized CNN. For MLFFC, internal layer fusion is employed while in MMFFC, fusion is performed using two models. EfficientNetB7 is implemented with Compound Scaling that increases the scale of a CNN. The results are as follows: MLFFC achieved 99.69% accuracy on CK + and 70.29% on FER2013, MMFFC reached 92.04% on KDEF and 68.14% on FER2013, and EfficientNetB7 achieved 57.5% on FER2013. Later, Cheng et al. [26] presented two approaches that utilize ensembles of CNN classifiers for multiple-stage classification. The first approach assigns different training sets to each CNN, while the second changes the image size of the original data to create multiple training sets. Merging the results of the individual CNNs is done by majority voting. The two approaches are tested using the ORL database, which includes ten images of forty different subjects for a total of four hundred images. The first two models achieved 90.0% and 99.5%

accuracies respectively. Later Chantrapornchai et al. [27] assigned a model leveraging an AutoML framework which integrates SqueezeNet (a CNN design meant to minimize parameter count) and SE blocks with residual block merges. The authors suggest such combinations can outperform traditional models without tuning outperform traditional models lacking hand tuning. The datasets used are CIFAR-10 and Tsinghua Facial Expression. For FER, the proposed method attains up to 71% accuracy (with proper SE block insertion, correct fire module quantity, and appropriate input merging) compared to less than 20% accuracy achieved by traditional models. Also, Abdulrahman et al. [28] introduced a Vision transformer model integrated with an SE block. With this approach, novel attention mechanisms were provided while alleviating the lack of available training data. This model achieved 99.8% accuracy for CK + dataset, 92.92% for JAFFE, 87.22% for RAF-DB, and 54.29% for SFEW, thereby surpassing or meeting the SOTA on these datasets.

Additionally, the Swin-Fer network was proposed by Mei et al. in [29], built upon the Swin transformer which captures global contextual information by using a fusion strategy in the middle to deeper layers, and adopting data dimension conversion techniques to obtain more spatial information. Amid the innovation, they also implemented a new method with which they accomplished 100% accuracy on the CK + dataset and 71.11% accuracy on FER2013. Also, Feng et al. [30] in their work advanced further by proposing another method of fine-tuning the Swin transformer alongside Multiple Weights Optimality-seeking. This model surpassed the state-of-the-art results for RAF-DB, FERPlus, and AffectNet datasets. Tao and Duan [31] used hierarchical attention networks with progressive feature fusion for FER. A range of studies showed that the introduced model performed better than all the other FER methods. Naveen [32] used a combination of Hopfield Networks, Deep Belief Network (DBN) architecture, and Lanczos interpolation to identify obscured facial expressions. This paper greatly contributes the theory, practice, and future research on facial expression recognition by providing a robust and effective methodology. Zhang et al. [33] proposed the spatial channel cross-fusion dual-attention network. The spatial channel focuses the features while the channel dimension offers a wider receptive field. Mehendale [34] advanced facial emotion recognition (FER) using convolutional neural networks (CNN) by considering deep network and traditional machine learning approaches. The FERC obtains facial feature vectors by background extraction through a two-stage CNN. To address demographic biases in affective computing AI systems, Singhal et al. [35] put forth the Demographic Biased Mitigating Framework (DBMF). The criticism of AI-based automated FER systems for not considering multi-modal signals gave birth to Li et al. [36] FER-former, a transformer network for FER in unconstrained settings with supervision from other modalities. To tackle the issue of restricted receptive fields, a hybrid feature extraction pipeline that combines cascaded CNNs and transformers is employed. Ge et al. [37] later developed an occluded expression recognition model with a countermeasure network. The model consists of two modules: face recognition and restoration of the obstructed face image. Table 1 presents the comparison of different methods using FER 2023.

2.2 Limitations of literature

There are several limitations to the state-of-the-art, including that most of the proposed models are hand-tuned by professionals, which requires cutting-edge technology,

Table 1 Comparison of existing methods and limitations using FER2023

Author	Method	Description	Accuracy (%)	Limitation
Pham et al. [19]	RMN	CNN-based model with residual, masking layers	74.14	Hand-tuned; dataset specific; performance drop on unseen data
Li [24]	FER using CNN	CNNs for spatial features, LSTMs for temporal modelling, transfer learning to improve performance	67.7	Lower accuracy; complex architecture; resource intensive
Thacker [25]	MLFCC	Multi-layer feature fusion approach, internal CNN layers fused.	70.29	Less effective on in-the-wild datasets
	MMFFC	Feature fusion to combine outputs from multiple models	68.14	Complexity increases; not generalizable
	EfficientNetB7	Lightweight CNN with compound scaling	57.5	Low performance on FER2013; overfitting risk
Mei et al. [29]	Swin-FER	Transformer-based architecture using Swin Transformer blocks and fusion strategies	71.11	Requires high computation; may not perform well on small datasets

Fig. 1 Examples from the FER2013 dataset, left to right: Anger, Disgust, Fear, Happy, Neutral, Sad, Surprise

specific expertise in the subject matter and substantial resources. Even with these methods, most models perform poorly on in-the-wild datasets, which are the closest to replicating real-life scenarios. This creates a restriction which prevents the widespread adoption of FER models. AutoML can be used to overcome this barrier, as it takes away a significant portion of the work by automating model selection and tuning.

3 Proposed work

3.1 Dataset

There are two broad categories of datasets, the first of which are the controlled or lab datasets. These datasets are created in well-controlled environments like laboratories, with attention to lighting, the actors' posture, camera angles and the background. Examples include JAFFE, MMI, KDEF and CK+ [38]. The second type of dataset is known as in-the-wild datasets. These consist of images captured in real-world scenarios and are inherently challenging due to issues such as obstructions in the image, and variations in lighting, background and poses. Some well-known datasets which fall into this category are: FER2013, FER + and AffectNet [38–40]. These are harder to classify and will have a lower accuracy as compared to the controlled ones due to the challenges mentioned but are of great importance in the context of Big Data, as they most closely resemble the data which real-world FER models aim to predict by virtue of being created by gathering results on the internet.

FER2013 [11] is a widely used dataset, which consists of 35,887 greyscale (48×48 pixel) images divided into seven broad expressions: anger, disgust, fear, happy, neutral, sad and surprise. FER2013 dataset was originally used in the ICML 2013 Challenges in Representation Learning and included uncontrolled images. Figure 1 reflects an instance of the FER 2013 dataset. It is divided into two parts, training data (consisting of 28,709

images) and test data (7,178 images). This allows for an unbiased evaluation of the proposed models. The test set was separated before training to ensure that no leakage occurs, which would affect the metrics during evaluation. The images in the FER2013 dataset have been pre-processed (by the authors of the dataset) in the following ways:

(i) Grayscale Conversion: The images are in grayscale, which reduces the computational complexity by eliminating the colour channels (RGB) and leaving a single intensity channel.
(ii) Resizing and Scaling: All the images are standardized to 48×48 pixels, which ensures uniformity across the dataset and reduces the computational burden during training.
(iii) Feature Extraction: The images have all been cropped such that only the relevant features remain.

FER2013 is well-known for its difficulty, being an in-the-wild dataset. Another reason for its difficulty is that it contains low resolution images in addition to those which may have occlusions. This creates a difficulty for the model during training but helps it adapt for the challenges faced when sorting through Big Data. Table 2 introduces the details of the dataset.

The "Percentage of the dataset (%)" column indicates the proportion of each emotion category relative to the entire FER2013 dataset (i.e., out of 35,887 images). This highlights the inherent class imbalance, with some classes like "Happy" comprising over 25% of the data, while others like "Disgust" make up less than 2%. Such imbalances can impact model training and evaluation, making metrics beyond accuracy, such as precision, recall, and F1-score, particularly important.

3.2 Models

To ensure a comprehensive and rigorous analysis, four distinct models were employed, each utilizing varied configurations tailored to address specific elements of the problem. The Default Model (Sect. 4.1) utilized a medium-quality preset, along with a dynamic selection of learning rate and batch size, using the Adam optimizer and a 10% validation holdout. The Low Resource Model (Sect. 4.2), optimized for CPU usage, employed a medium-quality preset with a higher learning rate for faster training. The Balanced Configuration Model (Sect. 4.3) sought to achieve a compromise between computational speed and output quality by using a high-quality preset, a moderate learning rate (0.001), a batch size of 32, and the Adam optimizer. Finally, the High-Quality Model (Sect. 4.4) prioritized accuracy, employing the highest quality preset, a low learning rate (0.0005), a batch size of 16, and the AdamW optimizer.

Table 2 Breakdown of datapoints in the dataset

Expression label	No. of images in training set	No. of images in testing set	Percentage of the dataset (%)
Anger	3995	958	13.80
Disgust	436	111	1.52
Fear	4097	1024	14.26
Happy	7215	1774	25.04
Neutral	4965	1233	17.27
Sad	4830	1247	16.93
Surprise	3171	958	11.50
	Total: 28,709	**Total: 7178**	**Total: 100**

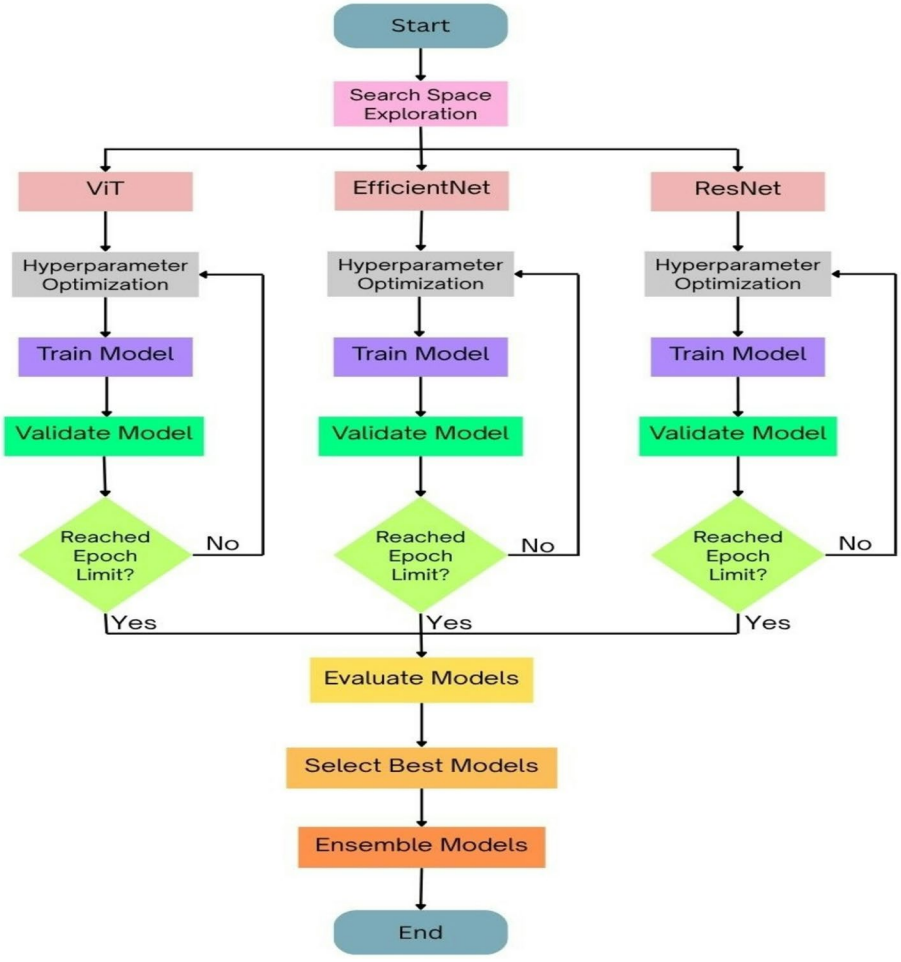

Fig. 2 Flowchart

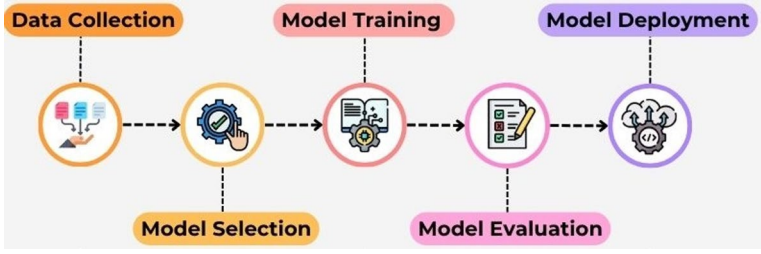

Fig. 3 Process diagram

3.3 Flowchart and process diagram

AutoGluon, the AutoML library used to create the model, uses a series of steps to find the model with the best results. The sequence is illustrated in the flowchart, Fig. 2 below, followed by a process diagram in Fig. 3 and the algorithm.

(i) **Search Space Exploration**: In this step, AutoGluon makes use of different base models from the PyTorch Image Models library like EfficientNet [41], ResNet [42], NFNet [43], ResNeXT [44], Vision Transformer (ViT) [45] and MobileNet [46] and

trains them on the dataset. This exploration phase ensures that the model considers diverse neural architectures for better coverage of learning patterns in the data.

(ii) **Hyperparameter Optimization**: Once the initial models are selected, AutoGluon employs optimization algorithms such as Bayesian optimization or random search algorithms, to search for the best configurations such as learning rate, number of layers, regularization strength etc. This step reduces the need for manual tuning and improves the model's ability to generalize well across varied input data.

(iii) **Train Models**: In this step, multiple models are trained using the prepared training data. During this phase, model weights are updated through multiple epochs. Each model is trained independently with its corresponding configuration derived from earlier steps.

(iv) **Validate Models**: After each epoch, the models are validated on a subset of the training data. This internal validation helps monitor the models' performance on unseen data and ensures the learning process creates a generalized model which is not overfitted.

(v) **Evaluate Models**: Once the models are trained, they are evaluated on a test set, which remains untouched during training to ensure no leakage of data. This final evaluation is used to assess the models' true generalization ability.

(vi) **Select Best Model**: The three best models are selected based on their evaluation performance.

(vii) **Ensemble Models**: In this step, the best fitted models are fused to further improve the accuracy of the final model using the formula presented in Eq. (1).

$$\hat{y} = g\left(f_1\left(x\right),\ f_2\left(x\right),\ f_3\left(x\right)\right) \tag{1}$$

where x is the input image, $f_1(x)$, $f_2(x)$, $f_3(x)$ are the predicted probabilities from the top three individual models and $g(x)$ is the meta-model (or ensemble function), which combines the individual predictions to produce the final output $\hat{y}$.

3.4 Algorithm

The following steps outline the process for training and evaluating models using AutoGluon, which includes converting datasets, initializing models, training, evaluating, and ensembling multiple models.

Algorithm: Training and Evaluation of Models in AutoGluon

Step 1: Convert raw training dataset into a dataframe with two columns: one for image locations and another for labels.
Step 2: Convert the test dataset into a dataframe in a similar format.
Step 3: Initialize MultiModalPredictor in AutoGluon, setting the label column to label.
Step 4: Start the model creation process by calling the fit() function with appropriate arguments for custom model configuration.
while Epoch limit not reached do
Step 5: Train the model on the training dataset.
end while
Step 6: Evaluate the trained model using the test dataframe.
if model accuracy is greater than the top 3 models then
Step 7: Save the model and discard the model with the lowest accuracy.
else
Discard the model being evaluated and re-run the training process with modified hyperparameters.
end if
Step 8: Ensemble the top three models to reduce overfitting and improve generalization

The raw data is converted into a structured dataframe that organizes image paths and corresponding labels. This is done for both training and test datasets. AutoGluon's MultiModalPredictor is initialized, focusing on the label column, which contains the class labels for classification tasks. In the training process, the model is trained iteratively, with each epoch updating model parameters. After training, the model is evaluated on the test dataset, and only models with performance higher than the top three models are retained. Thereafter, top-performing models are combined to mitigate overfitting and improve prediction performance.

3.5 Evaluation metrics

The proposed model is evaluated on different metrics like Accuracy, Precision, Recall and F-1 score. Accuracy is the main metric for measuring the efficacy of the proposed model, but it alone cannot provide an unbiased evaluation since FER2013 dataset is inherently unbalanced. It is apparent that there is a significant difference in the number of images in each class. For example, class "Disgust" has only 436 images in the training set while "Happy" has 7,215. Thus, using just accuracy to measure the efficacy of the resulting models may be misleading, so, for the purpose of this research, several metrics are used:

(i) **Accuracy**

Accuracy is the ratio of correctly identified instances (both true positives and true negatives) to the total number of instances (total images in the class). It is determined by the using the formula presented in Eq. (2). It is useful as it gives a quick overview of the how well the models perform across all classes and provides insights into the overall effectiveness of the models. However, it can be misleading due to the imbalanced nature of the FER2013 dataset, so a drawback of using just accuracy as a metric is that the model may predict the majority classes well and ignore the minority ones while still maintaining a high accuracy score.

$$Accuracy = \frac{Number\ of\ correct\ predictions}{Total\ number\ of\ predictions} \tag{2}$$

(ii) **Precision**

It is the ratio of true positives to the sum of the true positives and false positives. It is determined by the formula illustrated in Eq. (3). This metric is useful in determining how often a model is correct when it predicts an image to be in a certain class, for example, if it classifies an image to "Anger," how likely is it to be correct. This is important in situations where false positives are costly, for example in security where a model might misclassify "Neutral" as "Anger," thus creating suspicion and distrust.

$$Precision = \frac{Number\ of\ true\ positives}{Sum\ of\ true\ positives\ and\ false\ positives} \tag{3}$$

(iii) **Recall**

It measures of the model's ability to correctly identify all relevant instances of the given class as represented in Eq. (4). This metric helps identify the blind spots of a model, where it is failing to recognize an emotion reliably which is important for high-stakes

situations. For example, in healthcare, having a model with low recall for "Fear" or "Disgust" might mean that the system fails to recognize signs of genuine distress.

$$Recall = \frac{Number\ of\ true\ positives}{Sum\ of\ true\ positives\ and\ false\ negatives} \tag{4}$$

(iv) **F1 score**

It is the harmonic mean of precision and recall, determined by the formula presented in Eq. (5). This metric provides a single, balanced metric that recognizes a model's ability to avoid false positives (precision) and catch true positives (recall). It is especially useful in imbalanced datasets such as FER2013, where accuracy is deceptive.

$$\frac{1}{F1\ score} = \frac{1}{Precision} + \frac{1}{Recall} \tag{5}$$

4 Experiments and discussion

This section covers the experimentation and their subsequent analysis and discussion. Python version 3.12.2 and Autogluon version 1.2.0 are used for implementation. Experimentations are done using different models on FER 2013 dataset.

During training, after each epoch, the model was cross validated to ensure that the model performed well on unseen data. The cross-validation error formula is presented in Eq. (6).

$$CV\ \ Error = \frac{1}{k}\sum Error_i \tag{6}$$

where k is the number of folds the training set is divided into for validation and $Error_i$ is the error measured on the ith validation fold. In this case, the training dataset is divided into ten folds, such that nine are used for training and one for validation. Since every fold is used at least once as both training and validation during the learning phase, the performance is averaged across the divisions to account for the randomness of the split.

It is to be noted that no explicit feature selection or dimensionality reduction was performed by the authors during the training and preparation phase. However, AutoGluon performs implicit feature selection and dimensionality reduction through deep learning models like CNNs and ViTs via operations like convolution, pooling and attention, which enable the system to focus on the most relevant spatial features in the image data. Explicit parameters such as number of layers in the models or activation functions used were also not programmed in order to maintain the core objective of this work, which is to evaluate the performance of AutoGluon's automated configuration capabilities.

4.1 Default model

Since this is the baseline model for evaluation, no modifications are made to the parameters. For the default model, the medium-quality preset was employed, which balances between quality and speed, a dynamic learning rate (determined based on the gradient), automatically selected batch size (according to available resources ranging between 32 and 64), the Adam optimizer and a holdout of 10%, selected on the basis of the dataset. The medium quality preset is a hyperparameter optimization technique which defines the complexity of the model. In the case of the medium quality preset in the

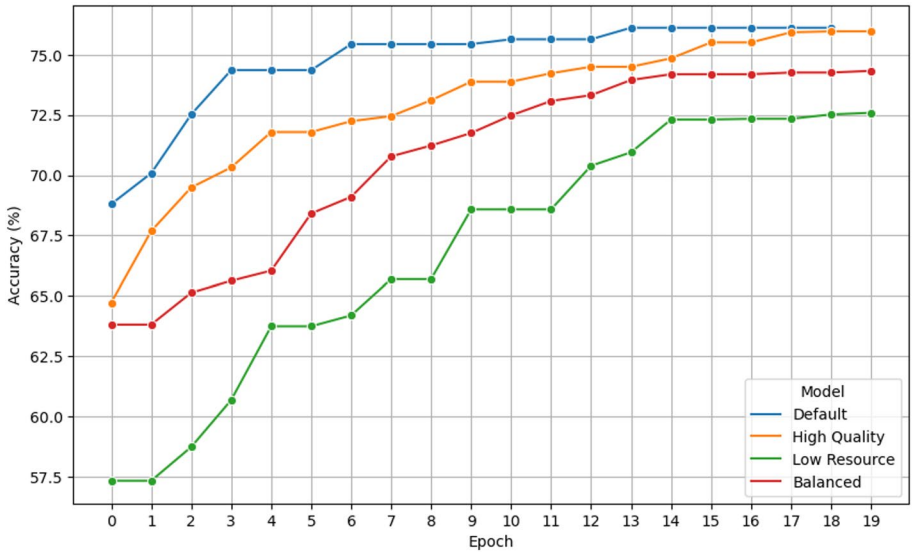

Fig. 4 Improvement in accuracy for proposed model through the epochs

Table 3 Result of default model

Class	Precision	Recall	F1-score	Support
Angry	0.73	0.67	0.70	958
Disgust	0.74	0.80	0.77	111
Fear	0.66	0.59	0.62	1024
Happy	0.92	0.93	0.93	1774
Neutral	0.72	0.75	0.74	1233
Sad	0.63	0.68	0.65	1247
Surprise	0.87	0.87	0.87	831
			Total	**7178**

Table 4 Confusion matrix for default model

Class	Angry	Disgust	Fear	Happy	Neutral	Sad	Surprise
Angry	616	16	94	28	71	115	18
Disgust	12	90	4	1	0	4	0
Fear	80	6	575	28	72	200	63
Happy	15	0	22	1612	76	21	28
Neutral	34	3	40	66	898	182	10
Sad	98	3	122	42	157	815	10
Surprise	18	0	44	30	17	15	707

default configuration, relatively fewer parameters were initialized, in this case 95.7 million. The model was trained to completion. In the default configuration, there is a Limit of 20 epochs. However, there was no improvement in the last five epochs, so the model terminated at epoch 19, also presented in Fig. 4. The training time was 4 h and the best accuracy during training was 76.12% while the test accuracy stood at 76.4%. Cases of the test accuracy being significantly lower than training accuracy are deemed to be a sign of overfitting. In this case, there was no overfitting. The model was also tested on precision, recall and F1-score and the respective values are presented in Table 3. Table 4 refers to the confusion matrix of default model. These results show that the default model

Table 5 Result for low resource model

Class	Precision	Recall	F1-score	Support
Angry	0.63	0.64	0.63	958
Disgust	0.73	0.67	0.70	111
Fear	0.58	0.56	0.57	1024
Happy	0.90	0.90	0.90	1774
Neutral	0.65	0.67	0.66	1233
Sad	0.59	0.59	0.59	1247
Surprise	0.84	0.82	0.83	831
			Total	**7178**

Table 6 Confusion matrix for low resource model

Class	Angry	Disgust	Fear	Happy	Neutral	Sad	Surprise
Angry	586	14	92	46	77	131	12
Disgust	23	77	8	1	0	2	0
Fear	101	6	544	29	96	186	62
Happy	33	0	26	1552	88	34	41
Neutral	82	2	77	78	794	182	18
Sad	121	4	156	47	195	707	17
Surprise	28	0	80	32	15	11	665

Generalizes well and reports an accuracy of 76.4%. It also showcases strong performance across most emotions except for "Fear" and "Sad" and does exceptionally well in "Happy."

4.2 Low resource model

This configuration was designed to evaluate how a model trained on a computer with Limited computing resources performs. Some people may not have access to high end GPUs required for traditional FER models and thus this model predicts how AutoGluon will perform when used on a CPU with low resources. The parameters selected for this model were a medium preset which is the lowest configuration possible to reduce the computational complexity, a relatively high learning rate of 0.01 to reduce training time, a batch size of 32 to reduce the number of batches per epoch and a Limit of 20 epochs. With the medium preset and low configuration settings, a model of only 4.2 million parameters was initialized. The holdout is 10%. This configuration prioritizes speed over performance, but the higher learning rate means the algorithm may overshoot the optimal solution and lead to a suboptimal model. Experimental results and confusion matrix are presented in Tables 5 and 6, respectively. These results revealed that the training time was approximately 40 min and at the end of 20 epochs, the best accuracy was 72.59%. After evaluation on the test set, the accuracy was 70.0%, lower than the other models. Similar accuracies on both sets indicate there was no overfitting. It's performance, remained like the default model, weakest in "Fear" and "Sad" emotions while doing well in predicting "Happy."

4.3 Balanced configuration model

This configuration was created to balance speed as well as quality. It offers better accuracy than the low resource configuration while demanding less resources than a model with emphasis on quality. For this purpose, it used the high-quality preset and has a moderate learning rate of 0.001, both which are between those of the low resource and high-quality models. The high-quality preset ensures that the model has an increased

Table 7 Results for balanced configuration model

Class	Precision	Recall	F1-score	Support
Angry	0.67	0.69	0.68	958
Disgust	0.73	0.73	0.73	111
Fear	0.63	0.59	0.61	1024
Happy	0.92	0.92	0.92	1774
Neutral	0.69	0.70	0.70	1233
Sad	0.63	0.63	0.63	1247
Surprise	0.85	0.87	0.86	831
			Total	**7178**

Table 8 Confusion matrix for balanced configuration model

Class	Angry	Disgust	Fear	Happy	Neutral	Sad	Surprise
Angry	639	17	89	27	61	107	18
Disgust	20	85	2	1	0	5	0
Fear	93	1	574	21	83	189	62
Happy	23	3	20	1588	82	22	36
Neutral	71	3	58	64	837	181	19
Sad	128	3	148	36	179	740	13
Surprise	19	0	57	34	11	9	701

complexity and can make deeper correlations in the data. In this case, with the high-quality preset and balanced configuration, it had a model size of 95.7 million parameters, the same as that of the default model. Thus, it is inferred that a medium quality preset with default configuration makes up for a high-quality preset. It has a batch size of 32 and uses the Adam optimizer. The holdout for cross-validation is 10%. The experimental results are illustrated in Table 7 while Table 8 presented the confusion matrix. The training time was 3 h, and accuracy during training was 74.33% while during testing it was 74.4% with no overfitting. The model did moderately well, performing close to the high-quality model and better than the low resource model. It too had difficulty differentiating between "Fear" and "Sad," thus the low precision scores but performed very well on "Happy."

4.4 High quality configuration

This configuration was optimized for quality over speed and used all the parameters necessary to reduce over-fitting and increase accuracy. It requires a high-end GPU and ample computing power to run. It used the best quality preset which prioritizes quality at the cost of speed, a low learning rate of 0.0005 which adjusts the model by small variations which helps it reach the optimal solution. Since this model used the best quality preset and highly optimized configurations for training, the model size was more than double any other model, with 195 million parameters. The batch size of 16 allowed for improved regularization by introducing more noise. Also, a smaller batch meant for more iterations per epoch, allowing the model to see more variations in the data sooner. Finally, the Adamw optimizer was used to keep the model from overfitting with its weight decay regularization. The training time was just over 7 h. The accuracy after 20 epochs in training was 75.97%, and in testing was 75.1%, trailing slightly behind the default model despite being fine-tuned for performance. The precision, recall and F1-scores are presented in Tables 9 and 10 shows the confusion matrix. The high-quality

Table 9 Results for high quality model

Class	Precision	Recall	F1-score	Support
Angry	0.67	0.69	0.68	958
Disgust	0.73	0.73	0.73	111
Fear	0.63	0.59	0.61	1024
Happy	0.92	0.92	0.92	1774
Neutral	0.69	0.70	0.70	1233
Sad	0.63	0.63	0.63	1247
Surprise	0.85	0.87	0.86	831
			Total	**7178**

Table 10 Confusion matrix for high quality model

Class	Angry	Disgust	Fear	Happy	Neutral	Sad	Surprise
Angry	645	9	97	26	72	97	12
Disgust	16	85	3	0	2	5	0
Fear	112	4	573	16	76	162	81
Happy	30	0	17	1578	90	24	35
Neutral	54	0	54	65	883	164	13
Sad	125	3	125	39	184	749	22
Surprise	17	0	53	30	13	15	703

Table 11 Comparison of the hyperparameters between automl models

Model	Learning rate	Batch size	Optimizer	Preset
Default	Dynamic	Auto (32)	Adam	Medium
Low resource	0.01	32	Adam	Medium
Balanced	0.001	32	Adam	High quality
High quality	0.0005	16	Adamw	Best quality

model did not perform noticeably better than the default model despite its highly tuned configuration. It performed slightly better only in the minority class "Disgust."

4.5 Comparison between different automl models

Table 11, compares the learning rate, batch size, optimizer, and presets for all the different models and broadly presents the comparison of hyperparameters among all the models.

There are some observations that are common to all the models. The prediction for "Happy" consistently performed well for all the models. This can be due to the large number of images (25% of the dataset) in that category as well as the uniqueness of the expression as compared to the others. The models show an above-average score for precision for "Disgust" despite it only comprising 1.52% of the dataset which is most likely due to the uniqueness of the emotion (Tables 3, 5, 7 and 9). Critically reviewing the confusion matrices (Tables 4, 6, 8 and 10) shows that the models mainly misattribute "Disgust" to "Anger" and vice versa owing to similar traits in the expressions. Similarly, "Fear," "Anger" and "Sad" share similar traits in their expressions and thus can be misattributed consistently as seen in Tables 4, 6 and 8, and 10. This can only be remedied by changing the way the foundational models distinguish one expression from another. The relatively lower performance in precision as compared to the state-of-the-art accuracy highlights the importance of the addition of other metrics in the evaluation criteria for the models.

Table 12 Comparison of automl models

Auto ML models	Training time (hours)	Accuracy	Precision (macro average)	Recall (macro average)	F1-score (macro average)
Default	4	76.4%	75%	76%	75%
Low resource	0.66	70%	69%	68%	69%
Balanced	3	74.4%	73%	73%	73%
High quality	7	75.1%	75%	74%	74%

Table 13 Comparison of state-of-the-art (SOTA) with different proposed models

Author	Method	Accuracy (%)
Pham et al. [19]	RMN	74.14
Li [24]	FER using CNN	67.7
Thacker [25]	MLFCC	70.29
	MMFFC	68.14
	EfficientNetB7	57.5
Mei et al. [29]	Swin-FER	71.11
Proposed model	**Default model**	**76.4**
	High quality model	75.1
	Balanced model	74.4
	Low resource model	70.0

Since "Happy" has such a high weightage in the dataset, it is correctly predicted much more often than the others and tends to skew the accuracy to provide a biased answer.

It is also interesting to note that despite the significantly higher training time and computational complexity, the higher quality model did not outperform the baseline default model presented in Fig. 4. It took 42% longer to train yet could not provide better results. However, where the default model stopped improving, the higher quality model showed small yet consistent improvements, which, given sufficient training time might lead to it outperforming the baseline. Experimental results presented in Table 12, it is apparent that the default model is better suited for Facial Expression Recognition tasks even compared to the more finely tuned models. The high-quality model provides slightly better performance in minority classes, but the value is minimal when compared to the increase in training time and complexity. So, the default model is best suited when resources are available. The balanced model is a good compromise for users who need a faster training time without sacrificing too much performance. The low-resource model can be used when computing power is limited but accuracy is not critical.

Figure 4 shows the difference the learning rates and batch sizes make. The low resource model, having the highest learning rate, began with a very low accuracy score, then showed considerable improvement after each epoch. The default model achieves the best performance across all epochs, reaching an accuracy of over 75% by the seventh epoch. This shows that the default configuration is well-tuned for FER tasks, balancing efficiency with high performance. The balanced model achieved a final accuracy only slightly behind the default model while taking less time and resources to train. This shows that it is the optimal model for non-critical FER tasks.

4.6 Comparison of different state-of-the-art with automl models

The performance of various state-of-the-art FER models was evaluated against the proposed models in terms of accuracy and the results are presented in Table 13. The

proposed models, particularly default model, demonstrate improved accuracy over conventional models.

The experimental results illustrated in Table 13 clearly states that RMN method, achieving an accuracy of 74.14%, is the highest among the SOTA, but was outperformed by the default, high quality and balanced models in the proposed approach. Li [24] used CNNs for FER, resulting in a lower accuracy of 67.7%, which was outperformed by all the models in the proposed approach. This shows that CNN by itself is not ideal for this dataset. Thacker [25] explored three different techniques, among which MLFCC had the highest accuracy at 70.29%, followed by MMFFC at 68.14% and EfficientNetB7 at 57.5%. Swin-FER [29] had an accuracy of 71.11%, outperformed by the default, balanced and high-quality model. Overall, the proposed approach delivered the highest accuracy. The default model achieved 76.4%, outperforming all SOTA methods, while the high-quality model reached 75.1%, also surpassing previous benchmarks. The balanced model showed a solid performance at 74.4%, matching the RMN method [19]. The low-resource model, designed for environments with Limited computational power, reached 70.0%, demonstrating competitive performance against the MLFCC method [25].

4.7 Comparative analysis with other automl frameworks

In comparison to other AutoML frameworks, AutoGluon is specifically designed for multimodal problems and supports deep learning frameworks, making it well-suited for computer vision tasks like FER. In contrast, other frameworks primarily focus on tabular data using genetic programming and are not optimized for image-based tasks unless pre-processed into numerical features. Similarly, Google Cloud AutoML Vision provides a no-code interface for image classification but offers limited transparency and control over model architecture, hyperparameters, and training strategy.

In terms of accuracy, AutoGluon's default configuration achieved 76.4% on the FER2013 dataset, outperforming state-of-the-art manually tuned models. While comparative FER results from TPOT are limited due to its focus on structured data, Google AutoML Vision has shown competitive performance in general image classification but lacks published FER benchmarks on FER2013. In terms of training time and resource requirements, AutoGluon provides flexibility through different presets (e.g., Low Resource, Balanced, High Quality), allowing users to balance performance with computational cost. Google Cloud AutoML abstracts away resource configuration but often requires longer cloud-based training cycles and incurs usage costs. TPOT, although lightweight, is less suited for deep image models and typically requires external feature engineering for FER. Overall, AutoGluon strikes a strong balance between performance, control, and automation—making it particularly advantageous for FER tasks that require deep image model customization and rapid experimentation.

4.8 Discussion

This paper demonstrated the practical and technical viability of using AutoML, particularly AutoGluon, for facial expression recognition (FER). Among the four proposed configurations, the Default Model achieved the highest test accuracy (76.4%), outperforming both traditional FER methods and more complex AutoML configurations. This suggests that AutoML can autonomously discover efficient architectures and hyperparameters without manual tuning.

Interestingly, the High-Quality Model, despite having significantly more parameters and longer training time, did not outperform the Default Model. This can be attributed to over-parameterization and slower convergence. These findings highlight the AutoML system's ability to prioritize practical, efficient configurations over sheer complexity. The results also affirm the ability of AutoGluon to manage imbalanced and noisy datasets like FER2013, which are notoriously difficult for many manually tuned models. By using automated hyperparameter optimization and ensembling, AutoGluon was able to generalize well across emotion categories.

4.9 Limitations

Despite its promising outcomes, this research has several limitations mentioned below:

(i) Computational Demand: AutoML still requires substantial computational resources, particularly for high-quality configurations. The high-quality model took over 7 h to train and may not be feasible for users with limited hardware.
(ii) Dataset Specificity: The models were trained and evaluated on FER2013. While it is a widely used benchmark, the generalizability of the results to other datasets such as AffectNet or SFEW remains untested in this paper.
(iii) Black-Box Nature of AutoML: While AutoGluon simplifies the pipeline for non-experts, it also abstracts internal processes, which may be a drawback for researchers who wish to understand or manually modify specific layers or operations.
(iv) Bias in Imbalanced Data: Despite the use of advanced metrics (precision, recall, F1-score), the performance remained weak for minority classes such as "Disgust," which constituted only 1.52% of the dataset. This imbalance could lead to biased predictions in real-world applications.
(v) Bias in Foundational Models: Due to the ensembling of pre-trained models, there is a high probability of perpetuation of inherent biases which may be present in the foundational models, which will lead to those same biases appearing in models created by AutoGluon.
(vi) Inability of Ablation Studies: Due to the automation of model selection and ensembling, it is not possible to do ablation studies on the proposed models which might reduce transparency of the model.
(vii) Absence of statistical validation: The proposed work is unable to conduct significance tests and obtaining confidence intervals due to resource constraints.

4.10 Future work

Future work can build on this research in the following ways:

(i) Cross-Dataset Generalization: To validate robustness, the trained models should be tested or fine-tuned on other FER datasets like AffectNet, FER+, or SFEW. This would help assess generalizability.
(ii) Model Interpretability: Incorporating explainable AI (XAI) methods can help interpret the decision-making of AutoML-generated models, increasing their trustworthiness in sensitive applications like healthcare or security.
(iii) Hybrid AutoML Strategies: Exploring hybrid pipelines that combine AutoML for model exploration with expert-guided fine-tuning could yield both accessible and highly optimized FER systems.

(iv) Deployment Optimization: Investigating lightweight AutoML models for mobile or embedded systems would expand the applicability of FER systems to edge devices where computational resources are limited.
(v) Bias Mitigation Techniques: Future work should address the dataset imbalance by integrating synthetic data augmentation, re-sampling techniques, or cost-sensitive learning to improve performance on underrepresented emotion classes.
(vi) Statistical validation: With the availability of high-end computing facility, significance tests can be conducted to obtain confidence intervals.

4.11 Ethical considerations in FER applications

The deployment of FER systems, particularly in sensitive domains such as healthcare, security, and education, raises critical ethical concerns. These include risks of misclassification, bias, privacy invasion, and over-reliance on automated decisions. To mitigate these issues, it is essential to ensure transparency, apply bias mitigation strategies, obtain informed consent, and position FER tools as assistive rather than authoritative systems. Addressing these ethical challenges is key to ensuring the responsible and equitable use of FER technologies in real-world applications.

4.12 Real-World applicability

The increasing demand for affective computing systems across domains such as healthcare, transportation, education, and public safety necessitates FER systems that are not only accurate but also deployable under real-world constraints. The proposed AutoML-based framework offers practical advantages, enabling its use across a range of deployment environments.

(i) Deployment Scenarios: The models can be deployed in the fields of healthcare monitoring, driver drowsiness detection, online education engagement and public surveillance.
(ii) Computational Complexity: The current work includes four configurations, (Default, Low Resource, Balanced and High Quality), each designed to work in different environments with different computational resources.
(iii) Real World Efficiency: A comparative analysis between the models in terms of training time, accuracy and performance metrics is shown in Sect. 4.5 and Table 12, which allows users to choose a model which fits their deployment constraints.
(iv) Deployment using AutoGluon: AutoGluon automates pipeline creation, thereby reducing the need for deep technical expertise and making the models suitable for non-expert deployment.

5 Conclusion

In this paper, AutoML was explored to find the best model for facial expression detection to overcome the challenges of traditional methods. Experimental results demonstrated that AutoML could effectively automate model selection, hyperparameter optimization and ensemble learning, significantly reducing the manual effort traditionally required for FER model development. These results also revealed that AutoML is a viable alternative to traditional methods of model creation, as it outperformed the conventional state-of-the-art during evaluation. It is also an effective method when dealing with the large volume and variety of images inherent in Big Data. This can have

significant implications for real-world applications of FER, including emotion analysis in large-scale market research, healthcare monitoring, real-time security and surveillance, educational engagement assessment and driver drowsiness detection. Experiments on the FER2013 dataset showed encouraging results, with the AutoML-generated models achieving SOTA accuracies of 76.4%, 70.0%, 74.4% and 75.1%.

Employing AutoGluon in FER offers transformative benefits that directly address long-standing challenges in model development. Traditional FER systems require extensive manual effort in model architecture design, hyperparameter tuning, and performance optimization—processes that are time-consuming, resource-intensive and demand expert knowledge. AutoGluon simplifies and accelerates these processes through automation, enabling non-experts to build high-performing models with minimal intervention. Specifically, AutoGluon enhances FER by dynamically exploring a diverse set of deep learning architectures (such as ResNet, ViT, and EfficientNet), fine-tuning them using advanced techniques like Bayesian optimization, and ensembling top-performing models to maximize generalization and accuracy. This not only reduces the risk of overfitting but also boosts performance on challenging, in-the-wild datasets like FER2013. As demonstrated in the paper, AutoGluon achieved state-of-the-art accuracy of 76.4%, outperforming several manually tuned models while significantly cutting down development time and computational cost. By democratizing access to robust FER systems, AutoGluon has the potential to accelerate real-world deployment in domains like healthcare, security, and emotion-aware applications, making intelligent emotion recognition more scalable, adaptable, and inclusive.

However, there were also certain limitations revealed during the research, including the dependency on the quality and diversity of the training data, the need for high computational power and the possibility of perpetuation of bias which may be present in foundational models. Future research could focus on addressing these limitations by exploring data augmentation techniques, using multi-modal data and distributed computing resources as well as making sure that the underlying data is scrutinized to prevent the reinforcement of inherent biases in the models. In the current times, where artificial intelligence and machine learning are being applied to every domain and have use cases in almost every application, there is an urgent need to allow non-experts the path to implement their own ML systems, a path that once remained blocked due to the skills and knowledge required to build these systems. AutoML is the key to enabling the ubiquitous use of ML technology and further integrate it into everyday applications, while also furthering the state-of-the-art.

Acknowledgements
Authors, thank the reviewers for their suggestions that have helped in improving this manuscript substantially. Authors also acknowledge Manipal University Jaipur for covering the APC under University agreement.

Author contributions
Atharv Goel: Conceptualizing, methodology, implementation, writing original draft.Praneet Saurabh: Conceptualizing, reviewing the manuscript and final drafting.Dhananjay Bisen: Conceptualizing, implementation and reviewing the final draft.Rishav Dubey: Conceptualizing, Supervising, Reviewing & final drafting.Prathap Somu: Reviewing & final drafting.

Funding
Open access funding provided by Manipal University Jaipur. The authors did not receive any funding to carry out the research presented in this manuscript.

Data availability
No datasets were generated or analysed during the current study.

Declarations

Ethical approval
This manuscript has followed all the ethical standards.

Consent for publish
Informed consents were obtained from all participants.

Consent to participate
Not applicable.

Competing interests
The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this manuscript.

Clinical trial number
Not applicable.

Received: 13 February 2025 / Accepted: 3 September 2025

Published online: 09 October 2025

References
1. Zhang HL, Lee S, Lu Y, Yu Xin, Lu Huanda. A survey on big data technologies and their applications to the metaverse: past, current and future. Mathematics. 2022;11:96. https://doi.org/10.3390/math11010096.
2. Kaur H, Kaur T, Bhardwaj V, et al. An ensemble deep learning model for classification of students as weak and strong learners via multiparametric analysis. Discov Appl Sci. 2024;6:595. https://doi.org/10.1007/s42452-024-06274-6.
3. Özbeyaz A, Yıldırım M, Tufaner F. Prediction of asthma outpatients using cumulative particulate matter and machine learning algorithms: a case study in adiyaman, Turkey. Discov Appl Sci. 2025;7:19. https://doi.org/10.1007/s42452-024-06407-x.
4. Atee M, Hoti K, Parsons R, Hughes J. Pain assessment in dementia: evaluation of a point-of-care technological solution. J Alzheimers Dis. 2017. https://doi.org/10.3233/JAD-170375.
5. Scientists use facial. recognition technology to develop tool for monitoring ICU patients' safety. News-Medical Life Sciences. [Internet]. 2019 [cited 2025 Jan 20]. Available from: https://www.news-medical.net/news/20190603/Scientists-use-facial-recognition-technology-to-develop-tool-for-monitoring-ICU-patients-safety.aspx
6. Saeed S, Shah AA, Ehsan MK, Amirzada MR, Mahmood A, Mezgebo T. Automated Facial Expression Recognition Framework using Deep learning. J Healthc Eng 2022. 2022. https://doi.org/10.1155/2022/5707930.
7. Wang Y, Shaoqi Y, Yang L, Wei S , Jing L, Yang C, Xinji M, Xiping H, Wenqiang Z, Zhongxue G. A Survey on Facial Expression Recognition of Static and Dynamic Emotions. 2024. 10.48550/arXiv.2408.15777.
8. Jack RE, Sun W, Delis I, Garrod OG. Four not six: revealing culturally common facial expressions of emotion. J Exp Psychol Gen. 2016;145(6):708–30. https://doi.org/10.1037/xge0000162.
9. Orlando F, Salehin M, He A, Wang J et al. Apr., Automated Machine Learning: Past, Present and Future, Eng. Appl. Artif. Intell., 2024.
10. Tang Z, Fang H, Zhou S, Yang T, Zhong Z, Hu T et al. AutoGluon-Multimodal (AutoMM): supercharging multimodal automl with foundation models. Proceedings of the Third International Conference on Automated Machine Learning, PMLR 2024;256(15):1–35.
11. Sambare M. FER-2013 dataset [Internet]. Kaggle; 2025 [cited 2025 Jan 20]. Available from: https://www.kaggle.com/datasets/msambare/fer2013
12. Almazrouei M, Dweiri F, Aydin R, et al. A review on the advancements and challenges of artificial intelligence-based models for predictive maintenance of water injection pumps in the oil and gas industry. SN Appl Sci. 2023;5:391. https://doi.org/10.1007/s42452-023-05618-y.
13. Ramos L, Rivas-Echeverría F, Pérez AG, et al. Artificial intelligence and sustainability in the fashion industry: a review from 2010 to 2022. SN Appl Sci. 2023;5:387. https://doi.org/10.1007/s42452-023-05587-2.
14. Ortega-Martorell S, Bellfield RAA, Harrison S, et al. Mapping the global free expression landscape using machine learning. SN Appl Sci. 2023;5:354. https://doi.org/10.1007/s42452-023-05554-x.
15. Brynjolfsson E, McAfee A. The business of artificial intelligence. Harv Bus Rev. 2017;7:1–31.
16. Zareen SS, Hossain MS, Wang J, Kang Y. Recent innovations in machine learning for skin cancer lesion analysis and classification: a comprehensive analysis of computer-aided diagnosis. Precis Med Sci. 2025;14(1):15–40. https://doi.org/10.1002/prm2.12156.
17. Zareen SS, Sun G, Kundi M, Qadri SF, Qadri S. Enhancing skin cancer diagnosis with deep learning: a hybrid CNN-RNN approach. Comput Mater Continua. 2024;79(1):1497–519. https://doi.org/10.32604/cmc.2024.047418.
18. Zhang S, Zhang Y, Zhang Y, Wang Y, Song Z. A dual-direction attention mixed feature network for facial expression recognition. Electronics. 2023;12(17):3595.
19. Pham L, Vu TH, Tran TA. Facial expression recognition using residual masking network. 2020 25th int Conf on pattern recognition (ICPR). IEEE; 2021. pp. 4513–9. https://doi.org/10.1109/ICPR48806.2021.9411919.
20. Krizhevsky A, Sutskever I, Hinton GE. Imagenet classification with deep convolutional neural networks. Commun ACM. 2017;60(6):84–90. https://doi.org/10.1145/3065386.
21. Jain DK, Shamsolmoali P, Sehdev P. Extended deep neural network for facial emotion recognition. Pattern Recognit Lett. 2019;120:69–74. https://doi.org/10.1016/j.patrec.2018.12.002.
22. Lopes AT, Aguiar E, DeSouza AF, Oliveira-Santos T. Facial expression recognition with convolutional neural networks: coping with few data and the training sample order. Pattern Recognit. 2017;61:610–21. https://doi.org/10.1016/j.patcog.2016.07.026.

23. Wang Y, Lee Y, Song Y, Rong X. Facial expression recognition based on auxiliary models. Algorithms. 2019;12(11):227. https://doi.org/10.3390/a12110227.
24. Li T-HS, Kuo P-H, Tsai T-N, Luan P-C. CNN and LSTM based facial expression analysis model for a humanoid robot. IEEE Access. 2019;7:93998–4011. https://doi.org/10.1109/ACCESS.2019.2928364.
25. Thacker C. Ensemble of multi features for facial expression recognition using deep learning techniques [dissertation]. Shodhganga; 2021.
26. Cheng WC, Wu TY, Li DW. Ensemble convolutional neural networks for face recognition. In: Proceedings of the International Conference on Algorithms, Computing and Artificial Intelligence. ACM; 2018. pp. 1–6.
27. Chantrapornchai C, Kajkamhaeng S, Romphet P. Micro-architecture design exploration template for automl case study on SqueezeSEMAuto. Sci Rep. 2023;13:10642.
28. Abdulrahman M, Hammami W, Souici C, Khoudour K, Sebe R. Learning vision transformer with squeeze and excitation for facial expression recognition. ArXiv. 2021. https://doi.org/10.48550/ArXiv.2107.03107.
29. Mei B, Huan X, Yan G, Kai S, Xiangjiu C. Swin-FER: Swin transformer for facial expression recognition. Appl Sci. 2024;14(14):6125. https://doi.org/10.3390/app14146125.
30. Feng H, Huang W, Zhang D, Zhang B. Fine-tuning Swin transformer and multiple weights optimality-seeking for facial expression recognition. IEEE Access. 2023;11:9995–10003. https://doi.org/10.1109/ACCESS.2023.3237817.
31. Tao H, Duan Q. Hierarchical attention network with progressive feature fusion for facial expression recognition. Neural Netw. 2023. https://doi.org/10.1016/j.neunet.2023.11.033.
32. Naveen P. Occlusion-aware facial expression recognition: a deep learning approach. Multimed Tools Appl. 2024;83:32895–921. https://doi.org/10.1007/s11042-023-17013-1.
33. Zhang F, Chen G, Wang H, et al. Cf-dan: facial-expression recognition based on cross-fusion dual-attention network. Computational Visual Media. 2024;10:593–608. https://doi.org/10.1007/s41095-023-0369-x.
34. Mehendale N. Facial emotion recognition using convolutional neural networks (FERC). SN Appl Sci. 2020;2:446. https://doi.org/10.1007/s42452-020-2234-1.
35. Singhal P, Gokhale S, Shah A, et al. Domain adaptation for bias mitigation in affective computing: use cases for facial emotion recognition and sentiment analysis systems. Discov Appl Sci. 2025;7:229. https://doi.org/10.1007/s42452-025-06659-1
36. Li Y, Wang M, Gong M, Lu Y, Liu L. FER-former: multimodal transformer for facial expression recognition. IEEE Trans Multimedia. 2024. https://doi.org/10.1109/TMM.2024.3521788.
37. Ge H, Zhu Z, Dai Y, Wang B, Wu X. Facial expression recognition based on deep learning. Comput Methods Programs Biomed. 2022;215:106728.
38. Li S, Deng W. Deep facial expression recognition: a survey. IEEE Trans Affect Comput. 2022;13(3):1195–215. https://doi.org/10.1109/TAFFC.2020.2981446.
39. Kopalidis T, Solachidis V, Vretos N, Daras P. Advances in facial expression recognition: a survey of methods, benchmarks, models, and datasets. Information. 2024;15(3):135.
40. Guo X, Zhang Y, Lu S, et al. Facial expression recognition: A review. Multimed Tools Appl. 2024;83:23689–735. https://doi.org/10.1007/s11042-023-15982-x.
41. Koonce B. EfficientNet. Convolutional neural networks with swift for tensorflow. Berkeley, CA: A; 2021. https://doi.org/10.1007/978-1-4842-6168-2_10.
42. Shafiq M, Gu Z. Deep residual learning for image recognition: A survey. Appl Sci. 2022;12(18);1–43.
43. Brock A, De S, Smith S, Simonyan K. High-performance large-scale image recognition without normalization. ArXiv. 2021. https://doi.org/10.48550/ArXiv.2102.06171.
44. Mekruksavanich S, Jitpattanakul A. A deep learning network with aggregation residual transformation for human activity recognition using inertial and stretch sensors. Computers. 2023;12(7):Art. 141.
45. Dosovitskiy A, Beyer L, Kolesnikov A, et al. An image is worth 16x16 words: Transformers for image recognition at scale. ArXiv. 2020. https://doi.org/10.48550/ArXiv.2010.11929.
46. Mehta S, Agrawal R, Sharma V. An improved mobilenet architecture for real-time image classification on edge devices. Sensors (Basel). 2023;23(3):1265.

Publisher's note

Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.